International Immunopharmacology 26 (2015) 174–180

Contents lists available at ScienceDirect

International Immunopharmacology

journal homepage: www.elsevier.com/locate/intimp

# *Callicarpa japonica* Thunb. reduces inflammatory responses: A mouse model of lipopolysaccharide-induced acute lung injury

Na-Rae Shin [a,b], In-Sik Shin [a,c], Hyuk-Hwan Song [a], Ju-Mi Hong [a], Ok-Kyoung Kwon [a], Chan-Mi Jeon [a], Jung-Hee Kim [a], Sang-Woo Lee [d], Joong-Ku Lee [d], Hang Jin [e], Wan Yi Li [e], Sei-Ryang Oh [a], Kyu-Woung Hahn [b,*], Kyung-Seop Ahn [a,**]

[a] Natural Medicine Research Center, Korea Research Institute of Bioscience & Biotechnology, 30 Yeongudanji-ro, Ochang-eup, Cheongwon-gu, Chungju-si, Chungbuk 363-883, Republic of Korea
[b] Hannam University, 1646 Yuseong-daero, Yuseong-gu, Daejeon 305-811, Republic of Korea
[c] College of Veterinary Medicine, Chonnam National University, 77 Yongbong-ro, Buk-gu, Gwangju 500-757, Republic of Korea
[d] International Biological Material Research Center, Korea Research Institute of Bioscience and Bioscience & Biotechnology, 125 Gwahak-ro, Yuseong-gu, Daejeon 305-806, Republic of Korea
[e] Medicinal Plant Research Institute, Yunnan Academy of Agricultural Sciences (YAAS), Yunnan, China

ARTICLE INFO

*Article history:*
Received 3 November 2014
Received in revised form 21 January 2015
Accepted 27 January 2015
Available online 4 February 2015

*Keywords:*
*Callicarpa japonica* Thunb.
Acute lung injury
Inducible nitric oxide synthase
Interleukin-6

ABSTRACT

*Callicarpa japonica* Thunb. (CJT) is traditionally used as an herbal remedy for the treatment of inflammatory diseases. This study aimed to investigate the anti-inflammatory response of CJT in lipopolysaccharide (LPS)-stimulated RAW264.7 cells and LPS-induced acute lung injury (ALI) in mice. The C57BL/6 mice were administered 30 mg/kg of CJT by oral gavage for 3 days. LPS is applied to animals by intranasal administration 1 h after final CJT treatment. LPS is applied to animals by intranasal administration 1 h after final CJT treatment. LPS was delivered intranasally 1 h after the final CJT treatment. In the LPS-stimulated RAW264.7 cells, CJT significantly decreased nitric oxide (NO) and interleukin (IL)-6 in a concentration-dependent manner by reducing inducible NO synthase (iNOS) and IL-6 mRNA levels. In the ALI model, CJT decreased the inflammatory cell count in the bronchoalveolar lavage fluid (BALF) while IL-6 levels were reduced in CJT-treated mice compared with the ALI control mice. CJT also inhibited airway inflammation by reducing iNOS expression in lung tissue. In conclusion, our results indicate that CJT inhibits inflammatory responses in LPS-stimulated RAW264.7 cells and in the LPS-induced ALI model. Therefore, we suggest that CJT has the potential to treat inflammatory diseases such as pneumonia.



## 1. Introduction

Acute lung injury (ALI) is characterized by a sustained and uncontrolled inflammatory process in the lungs, resulting in airway dysfunction, involving neutrophil recruitment, interstitial edema, disruption of epithelial integrity, and lung parenchymal injury [1,2]. Clinical observations have indicated that Gram-negative bacterial infection plays an important role in the development of ALI. Lipopolysaccharide (LPS) is a major constituent of the outer membrane of Gram-negative bacteria and one of the most potent activators of the inflammatory response [3].

According to previous studies, mice with LPS-induced ALI demonstrated neutrophil, increased pro-inflammatory cytokine levels such as interleukin (IL)-6 and nitrogen species overproduction [4]. Nitric oxide (NO), a gas molecule that is synthesized by NO synthase (NOS), plays a crucial role in the development of inflammatory responses [5]. In *in vitro* experiments, macrophages were stimulated with agents such as LPS and the activated macrophages induced inflammation, which was accompanied by the production of pro-inflammatory molecules such as cytokines and NO [6]. Pro-inflammatory cytokines can eventually promote NO generation by inducible NOS (iNOS), resulting in its prolonged expression at higher concentrations, which activates a signal to promote inflammatory responses [7].

Despite recent advances in many new strategies for treating ALI, mortality due to ALI still remains greater than 40%. Thus, many researchers have investigated the effects of herbal-derived remedies on ALI [8,9]. Due to significant synergic therapeutic effects and relatively low toxicity of herbal medicine, considerable attention has been given to natural products with anti-oxidant and anti-inflammatory properties.

*Callicarpa japonica* Thunb. (CJT) is widely distributed in Korea and China, especially in temperate regions near the sea shore. Previous studies demonstrated that CJT has anti-inflammatory and anti-oxidant properties. According to Ono et al., CJT reduced reactive oxygen species

* Corresponding author. Tel.: +82 42 629 8757; fax: +82 42 629 8751.
** Correspondence to: K.-S. Ahn, Natural Medicine Research Center, Korea Research Institute of Bioscience and Biotechnology, 30 Yeongudanji-ro, Ochang-eup, Cheongwon-gu, Chungbuk 363-883, Republic of Korea. Tel.: +82 42 868 9464; fax: +82 42 864 2120.
*E-mail addresses:* hahnkw@hnu.kr (K.-W. Hahn), ksahn@kribb.re.kr (K.-S. Ahn).

http://dx.doi.org/10.1016/j.intimp.2015.01.025
1567-5769/

and free radical production [10]. However, no study has investigated CJT-mediated anti-inflammatory effects in a mouse model of ALI.

We investigated the inhibitory potential of CJT on inflammation in an ALI mouse model and LPS-stimulated RAW264.7 cells. To investigate the possible mechanism by which CJT reduced inflammation, we evaluated pro-inflammatory mediator expression and production in an ALI mouse model and in RAW264.7 cells.

## 2. Materials and methods

### 2.1. Materials

The murine macrophage cell line RAW264.7 was purchased from ATCC and CJT extract was obtained from The Plant Extract Bank at the Korea Research Institute of Bioscience and Biotechnology (KRIBB, FBM 161-088). CJT belongs to the Verbenaceae family and is collected from Beijing in China. The 3-(4,5-dimethylthiazol-2-yl)02,5-diphenyltetrazolium bromide (MTT, AMRESCO, OH, USA), Griess reagent (Promega Corporation, WI, USA) and enzyme-linked immunosorbent assay (ELISA) kit of IL-6 (R&D system, MN, USA) were used according to the manufacturer's instructions. In the case of the antibodies, anti-β-actin and -iNOS antibodies were obtained from Cell Signaling (MA, USA) and Enzo (MA, USA), respectively. LPS and dexamethasone were purchased from Sigma-Aldrich (CO, USA) and the Diff-Quik stain kit was obtained from IMEB (CA, USA).

### 2.2. UPLC Q-TOF-MS analysis

A UPLC was performed using an ACQUITY UPLCTM system (Waters Corporation, Milford, MA, USA) with photodiode array (PDA) detector and a sample manger coupled to a Micromass QTOF PremierTM mass spectrometer (Waters Corporation, Milford, MA) equipped with an electrospray interface. Chromatographic separations were performed on a 2.1 × 100 mm, 1.7 μm ACQUITY BEH C18 (Waters) chromatography column. The column temperature was maintained at 35 °C, and the mobile phases A and B were water with 0.1% formic acid and acetonitrile with 0.1% formic acid, respectively. The gradient conditions were as follows: 0 min, 10% B; 0–7 min, 10–33% B; 7–14 min, 33–56% B; 14–21 min, 56–100% B; 21–23 min, 100% B, and then held for 2 min before returning to the initial conditions. The flow rate was 0.4 ml/min, and the loading volume was 1.0 μl. The mass spectrometer was operated in a negative ion mode. N2 was used as the desolvation gas. The desolvation temperature was set to 350 °C at 400 l/h with a source temperature of 110 °C. The capillary and cone voltages were set to 2350 and 50 V, respectively. The data were collected for each sample with a 0.25 s scan time and a 0.01 s interscan delay. Leucine-enkephalin (*m*/*z* 554.2661) was used as a reference compound.

### 2.3. In vitro experiment

#### 2.3.1. Cell culture and cell viability

RAW264.7 was maintained in DMEM supplemented with 10% heat-inactivated fetal bovine serum (FBS) and antibiotics at 37 °C in a 5% $CO_2$ incubator. The cell viability after CJT treatment was measured using a MTT assay. The RAW264.7 cells were cultured in 96-well plates at a density of $1 \times 10^4$ cells/well. CJT extract was added to each individual well at 5, 10, 20, or 40 μg/ml and then incubated for 24 h. MTT solution (10 μl) was added to each well, followed by incubation for 4 h at 37 °C. After incubation, 100 μl dimethyl sulfoxide (DMSO) was added to each well for the formazan solubilization. The optical density was measured at 570 nm using a microplate reader (Molecular Devices Inc., CA, USA).

#### 2.3.2. Determination of NO production

Cells ($5 \times 10^4$ cells/well) were seeded in 96-well plates in FBS-free DMEM, treated with different CJT concentrations (5, 10, 20, and 40 μg/ml) for 1 h, and incubated in the presence of LPS (0.5 μg/ml) for 24 h. Nitrite accumulation in the culture medium was measured using the Griess reagent. The absorbance at 540 nm was measured using a microplate reader (Bio-Rad Laboratories, CA, USA).

#### 2.3.3. Determination of pro-inflammatory cytokine levels

Cells were treated as described above, and IL-6 concentration in the culture medium was quantified using a competitive ELISA kit according to the manufacturer's instructions. The absorbance at 450 nm was measured in a microplate reader (Bio-Rad Laboratories). Absolute concentrations were obtained by running standard curves on identical ELISA plates.

#### 2.3.4. Reverse transcriptase-polymerase chain reaction (RT-PCR)

Total RNA was isolated using TRIzol™ Reagent (Life Technologies Corporation, CA, USA). For RT-PCR, single-strand cDNA was synthesized from 2 μg total RNA. The primer sequences used were as follows: iNOS, sense 5′-CAA GAG TTT GAC CAG AGG ACC-3′ and antisense 5′-TGG AAC CAC TCG TAC TTG GGA-3′ and β-actin, sense 5′-CGC TCA TTG CCG ATA GTG AT-3′ and antisense 5′-TGT TTG AGA CCT TCA ACA CC-3′. PCR products were fractionated by 1.5% agarose gel electrophoresis and stained with 5 μg/ml ethidium bromide.

### 2.4. In vivo experiment

#### 2.4.1. Animals

Specific pathogen-free male C57BL/6 mice (20–25 g, 6–8 weeks old) were purchased from Koatech Co. (Pyeongtaek, Korea). They were housed in groups of 7 under standard conditions (temperature 22 ± 2 °C, humidity 55 ± 5%, 12-h light/dark cycle) with food and water. All of the experimental procedures were approved by the Institutional Animal Care and Use Committee of the Korea Research Institute of Bioscience and Biotechnology.

#### 2.4.2. Induction of ALI in C57BL/6 mice and drug administration

Mice were randomly divided into the control and 3 treatment groups. CJT (30 mg/kg; distilled water dissolved with 0.5% carboxymethyl cellulose (CMC)) and dexamethasone (Dex; 3 mg/kg) were administered orally from day 0 to day 2. Mice were treated with 50 μl LPS (20 μg/per mouse) intranasally (i.n.) 1 h after the final CJT and dexamethasone treatment. Normal mice were given 50 μl PBS without LPS.

#### 2.4.3. Bronchoalveolar lavage fluid (BALF) and serum collection

BALF collection was performed using the method of Shin et al. [11]. To obtain the BALF, ice-cold PBS (0.7 ml) was infused into the lungs two times and withdrawn each time using a tracheal cannula (a total volume of 1.4 ml). The BALF differential cell count was performed using Diff-Quik® staining reagent according to the manufacturer's instructions. Blood was collected from the inferior vena cava and centrifuged for 10 min at 4 °C. The supernatant was stored at −70 °C.

#### 2.4.4. Measurement of cytokines in BALF

Pro-inflammatory cytokine (IL-6) in BALF was measured using ELISA kits according to the manufacturer's protocols. Plates were incubated for 10 min in the dark, and the absorbance was measured at 450 nm in a microplate reader (Bio-Rad Laboratories).

#### 2.4.5. Immunoblotting

Lung tissue was homogenized (1/10 w/v) using a homogenizer in a Tissue Lysis/Extraction reagent (Sigma-Aldrich) containing protease inhibitor cocktail (Sigma-Aldrich). To extract nuclear protein, lung tissue was homogenized using an extraction kit (Thermo, CA, USA). The protein concentration for each sample was determined using the Bradford reagent (Bio-Rad Laboratories). Equal amounts of total protein (30 μg) were resolved by 10% SDS-polyacrylamide gel electrophoresis and transferred to nitrocellulose membranes. Membranes were incubated with blocking solution (5% skim milk) followed by overnight incubation at 4 °C with the appropriate primary antibodies. The following primary antibodies and dilutions were used: anti-β-actin (1:2000 dilution) and anti-iNOS (1:1000 dilution). Blots were washed three times with Tris-buffered saline containing Tween 20 (TBST) and then incubated with a 1:10,000 dilution of horseradish peroxidase (HRP)-conjugated secondary antibody (Jackson Immuno Research, PA, USA) for 30 min at room temperature. Blots were again washed three times with TBST and developed using an enhanced chemiluminescence (ECL) kit (Thermo).

*2.4.6. Lung tissue histopathology*

Lung tissue was fixed in 4% (v/v) paraformaldehyde and embedded in paraffin, sectioned at 4-μm thickness and stained with H&E solution (Sigma-Aldrich) to estimate inflammation.

*2.5. Statistical analysis*

Data are expressed as the means ± standard deviation (SD). Statistical significance was determined using analysis of variance (ANOVA) followed by a multiple comparison test with Dunnett's adjustment. *P* values < 0.05 were considered to be significant.

## 3. Results

*3.1. Identification of major constituent of C. japonica Thunb. by UPLC QTOF-MS*

The methanol extract profile was obtained by UPLC Q-TOF-MS analysis. The PDA chromatogram was shown in Fig. 1A. Two major metabolites could be identified by a carefully interpreting the mass spectra including the experimental *m/z* values, the error (ppm), and molecular formula. According to the MS spectra (Fig. 1B), compounds A and B were revealed at *m/z* 623.1924 (calculated for $C_{29}H_{35}O_{15}$) and 755.2372 ($C_{34}H_{43}O_{19}$), and their high resolution mass analysis and literature data were consistent with those of actioside and forsythoside B, respectively [12].

*3.2. In vivo experiment*

*3.2.1. CJT decreases inflammatory cell and pro-inflammatory cytokine content in BALF*

Mice treated with LPS demonstrated a significant increase in the inflammatory cell content of the BALF compared with the normal control mice. In particular, the number of neutrophil in BALF is markedly increased in LPS-induced mice compared with the normal control mice; however, CJT-treated mice significantly reduced the number of neutrophil in BALF compared with the LPS-induced mice (Fig. 2A).

IL-6 levels in the BALF were significantly increased in the LPS-induced mice compared with normal controls. In contrast, CJT-treated mice had significantly decreased IL-6 levels compared with the LPS-induced mice (Fig. 2B).

*3.2.2. CJT treatment inhibits iNOS expression in lung tissue of LPS-induced mice*

The LPS-induced mice demonstrated a significant increase in iNOS expression in lung tissue compared with normal controls. Dexamethasone-treated mice displayed reduced iNOS expression compared with LPS-induced mice. In the CJT-treated mice, a meaningful reduction in iNOS expression was also observed compared with LPS-induced mice (Fig. 3).

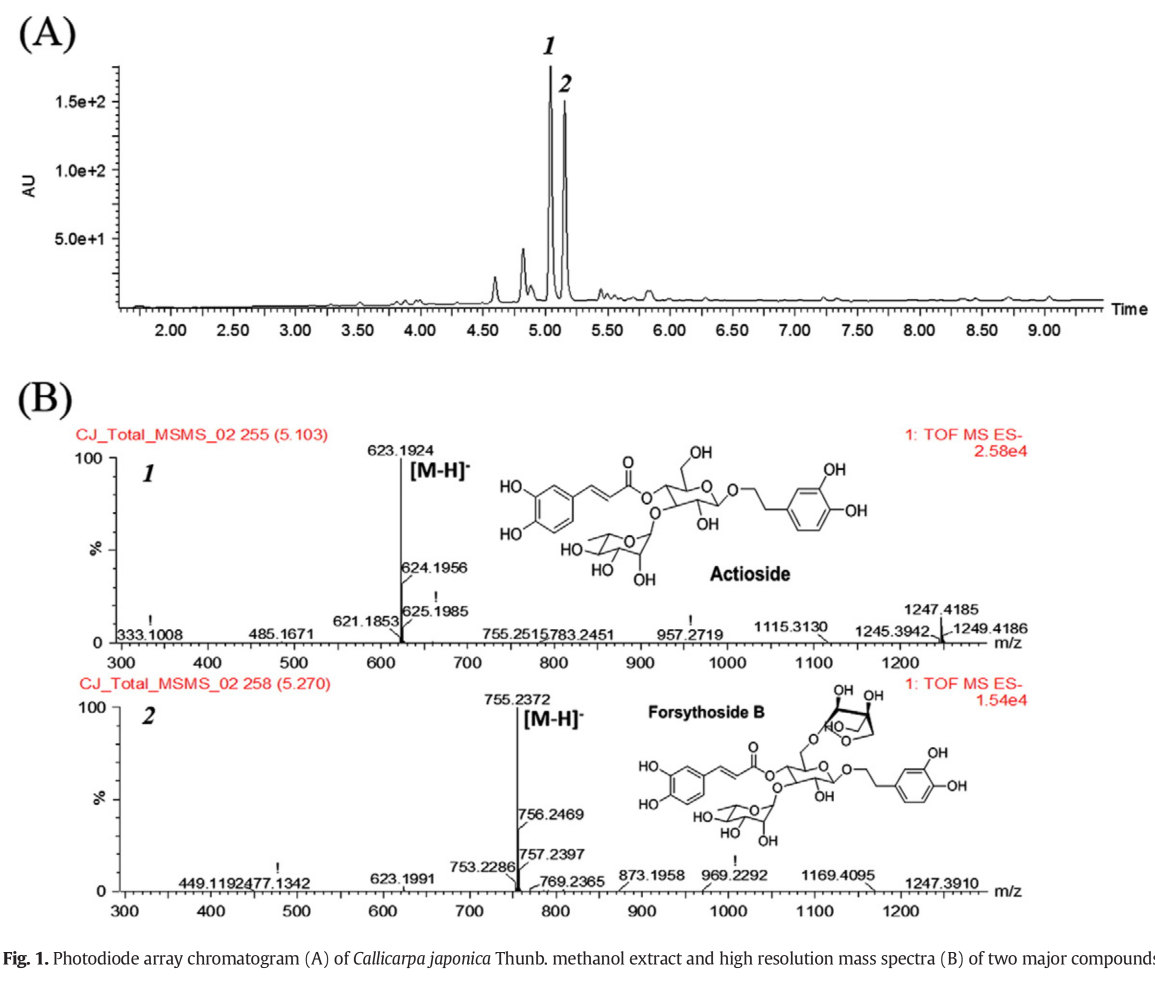


**Fig. 1.** Photodiode array chromatogram (A) of *Callicarpa japonica* Thunb. methanol extract and high resolution mass spectra (B) of two major compounds.

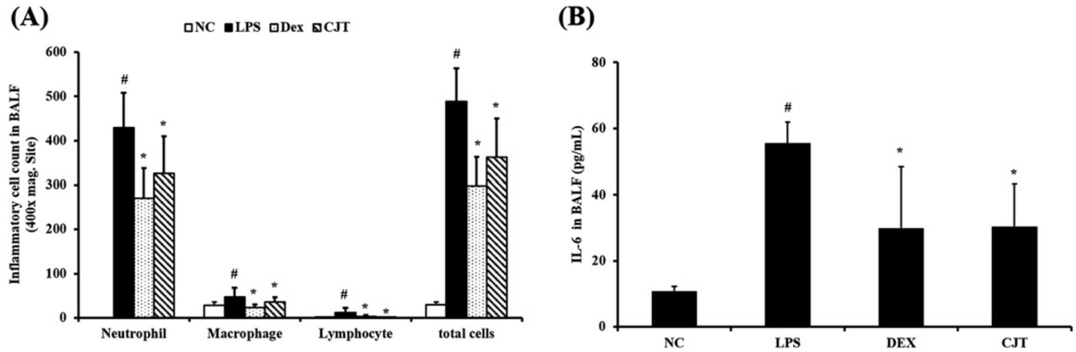


**Fig. 2.** CJT treatment inhibited inflammatory cell content and pro-inflammatory cytokine levels in mouse BALF (A). Cells were isolated by centrifugation and stained with Diff-Quik stain reagent. IL-6 levels were determined by ELISA (B). NC: normal control mice; LPS: LPS-induced mice; DEX: dexamethasone (3 mg/kg) + LPS-induced mice; CJT: CJT (30 mg/kg) + LPS-induced mice. The values are expressed as the means ± SD (n = 7/group). #Significantly different from NC, $P < 0.05$; *significantly different from LPS, $P < 0.05$.

*3.2.3. CJT treatment decreases inflammatory cell infiltration*

We observed a marked infiltration of inflammatory cells into the peribronchiolar and perivascular lesions in lung tissue sections from the LPS-induced mice compared with the normal mice (Fig. 4). Sections from dexamethasone-treated mice demonstrated mild inflammatory cell infiltration in the bronchial airway compared with LPS-induced mice. Similar to that of dexamethasone-treated mice, CJT-induced mice demonstrated reduced inflammatory cell infiltration compared with LPS-induced mice.

*3.3. In vitro experiment*

*3.3.1. CJT treatment reduces NO and IL-6 production in LPS-stimulated cells*

In this study, nontoxic CJT concentrations (5, 10, 20 and 40 μg/ml, Fig. 5A) were assessed for their ability to inhibit LPS-stimulated NO production. LPS-stimulated cells demonstrated increased cellular NO levels compared with non-stimulated cells. Cells pre-treated with CJT demonstrated reduced NO production in a concentration-dependent manner (Fig. 5B).

As demonstrated in Fig. 5C, LPS treatment markedly increased IL-6 levels compared with non-stimulated cells, whereas CJT-treated cells demonstrated significantly decreased IL-6 levels in a concentration-dependent manner compared with LPS-stimulated cells.

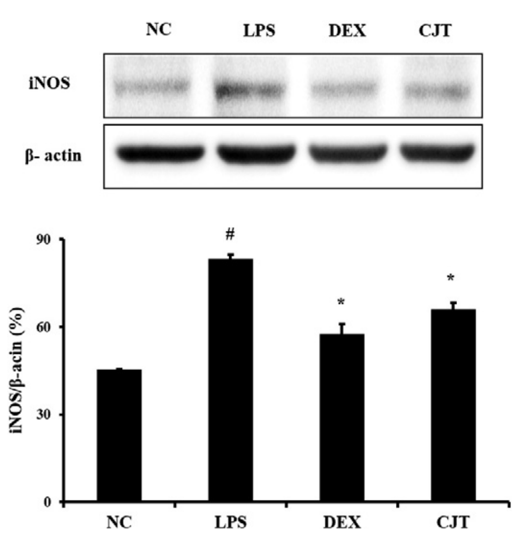


**Fig. 3.** CJT reduced iNOS protein levels in lung tissue. iNOS expression was assessed by Western blot analysis. β-Actin was used to confirm equal protein loading. NC: normal control mice; LPS: LPS-induced mice; DEX: dexamethasone (3 mg/kg) + LPS-induced mice; CJT: CJT (30 mg/kg) + LPS-induced mice. Values are expressed as the means ± SD (n = 7/group). #Significantly different from NC, $P < 0.05$; *significantly different from LPS, $P < 0.05$.

*3.3.2. CJT treatment inhibits iNOS mRNA expression in LPS-stimulated cells*

LPS-stimulated cells exhibited increased iNOS mRNA expression. However, treating cells with CJT significantly reduced iNOS mRNA expression compared with LPS-stimulated cells (Fig. 6). The ratio of iNOS mRNA expression was increased to a greater extent in CJT-treated cells than in LPS-stimulated cells.

## 4. Discussion

We evaluated the anti-inflammatory effect of CJT treatment on an LPS-induced ALI model and LPS-stimulated RAW264.7 cells. CJT treatment decreased inflammatory cell content and pro-inflammatory cytokine levels in BALF. CJT treatment also decreased iNOS expression levels as well as inflammatory cell infiltration in lung tissue. In LPS-stimulated RAW264.7 cells, CJT treatment inhibited NO production with reduced iNOS mRNA and pro-inflammatory cytokine expression.

Inflammation is a complex reaction against many pathological conditions including tissue injury [13]. The inflammatory response is induced by inflammatory mediators that are generated *via* a series of inducible genes that have critical functions in host immune defense, signal transduction pathways and vascular regulation [14]. Many stimuli can activate inflammatory cells such as neutrophils and macrophages, which are marked by secretion of pro-inflammatory mediators including IL-6 and NO near injury sites [13,15]. LPS, the major component of the outer membrane of Gram-negative bacteria, is an important risk factor for acute lung injury [16]. LPS acts as a causal factor in many serious infectious diseases such as sepsis, arthritis, and cancer [17]. In critical experimental mouse models of ALI, LPS administration injures both endothelial and epithelial barriers and indirectly activates neutrophils and macrophages, thereby releasing pro-inflammatory cytokines such as IL-6 [18,19]. IL-6 is a pro-inflammatory cytokine, and high IL-6 expression levels can promote the generation of active vascular substances, activate complement and neutrophils, promote neutrophils to cross the vascular endothelium and gather in tissue, and cause a cascade effect of inflammation to increase tissue injury [20]. In our study, CJT treatment reduced IL-6 levels in an ALI mouse model and in LPS-simulated RAW264.7 cells. Consistent with these results, CJT treatment significantly decreased the number of neutrophil in BALF and inhibited airway inflammatory cell infiltration into lung tissue. These findings indicate that CJT effectively inhibited inflammatory responses.

NO is a crucial inflammatory and neurotoxic mediator that increases pulmonary inflammation [21,22]. In LPS-induced ALI, large amounts of NO are generated *via* the up-regulation of iNOS, which is expressed in lung epithelium. iNOS contributes to the pathogenesis of murine acute lung injury models in a complex fashion with pro-inflammatory effects [23–25]. Previous studies have published results in which they demonstrated that iNOS knockout mice have less lung inflammation compared with wild-type mice [26]. Speyer et al. [27] and D'Alessio et al. [28]

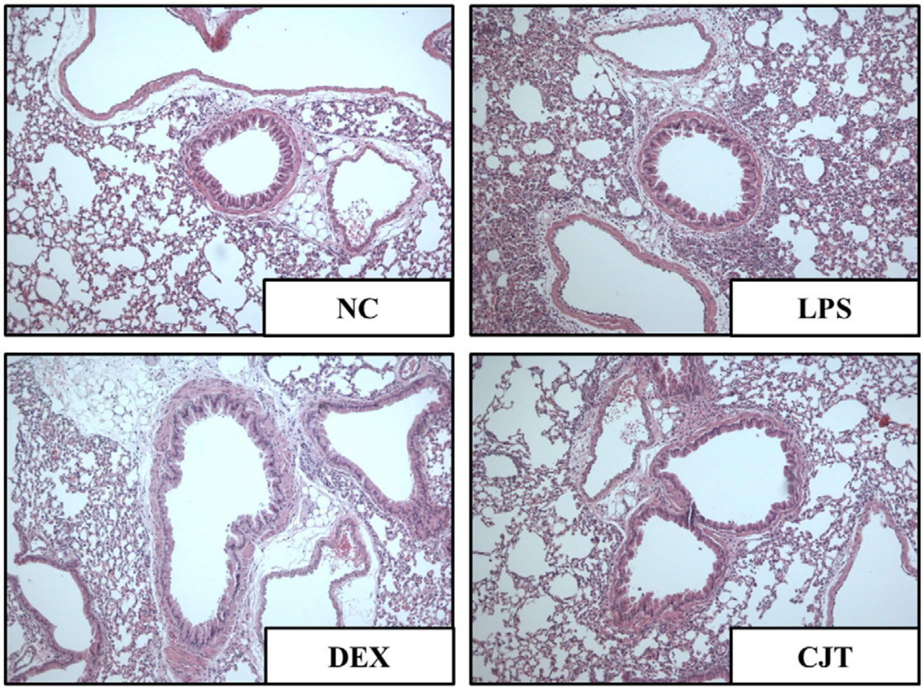


**Fig. 4.** CJT inhibited airway inflammation. Histological examination of airway inflammation performed in lung tissue by H&E staining. NC: normal control mice; LPS: LPS-induced mice; DEX: dexamethasone (3 mg/kg) + LPS-induced mice; CJT: CJT (30 mg/kg) + LPS-induced mice. Values are expressed as the means ± SD (n = 7/group). #Significantly different from NC, $P < 0.05$; *significantly different from LPS, $P < 0.05$.

demonstrated reduced inflammatory responses in an LPS-induced acute lung injury model. Similarly, Wang et al. [29] suggested an inhibition of inflammation in an endotoxin-induced lung injury model. Thus, previous reports demonstrate that an important mediator of lung inflammation is iNOS expression. Anti-inflammatory effect of CJT is considered to be related to both ROS scavenging activity and suppression of iNOS. Excess ROS induce the proinflammatory mediators by activation of inflammatory signaling such as NF-κB, iNOS and MAPKs, which eventually aggravates inflammatory responses in damaged lesions [30]. Therefore, natural materials that possess ROS scavenging activity have a potential for treatment of inflammation. Indeed, several natural products have been developed as anti-inflammatory agents based on their ROS scavenging activity [31–33]. On the other hand, NO is a gaseous and short-lived free radical that is synthesized from the conversion of L-arginine to L-citrulline, which is catalyzed by NO synthases (NOS) [34]. NOS has three isoforms namely neuronal (nNOS), endothelial (eNOS), inducible isozyme (iNOS) [35]. iNOS can generate $O_2$•—, depending on substrate and cofactor availability and combined with $O_2$•— is formed reactive nitrogen species (RNS) ONOO– [36,37]. So, suppression of iNOS is considered to reduce the inflammatory response *via* inhibiting specific inflammatory signaling and RNS production. Based on this information, we considered that anti-inflammatory effects of CJT are displayed by synergic responses of ROS scavenging activity and suppression of iNOS. In this study, we determined that CJT treatment decreased nitrite content and inhibited iNOS expression in the LPS-induced ALI mouse model and LPS-stimulated RAW264.7 cells. CJT-treated mice demonstrated reduced iNOS expression in lung tissue compared with LPS-induced mice, and CJT-treated RAW264.7 cells demonstrated decreased NO production and iNOS mRNA expression compared with controls. These results indicate that CJT treatment effectively inhibited inflammatory responses *via* the suppression of iNOS expression.

Anti-inflammatory effects of CJT were considered to be related to their active components including actioside and forsythoside B. According to previous studies [38–41], actioside and forsythoside B isolated from CJT have an anti-inflammatory activities. Actioside effectively inhibited inflammatory responses *via* the reduction in the ERK and JNK expression in IL-1β-induced human vascular endothelial cells [38] or the decrease in p38 MAPK expression in LPS-induced RAW264.7 cells [39]. In addition, forsythoside B significantly reduced inflammatory mediators including IL-6 and NF-κB in *in vitro* or *in vivo* experiments [40]. Furthermore, Jiang et al. [41] demonstrated that forsythoside B reduced ischemia–reperfusion injury in rat *via* inhibiting the production

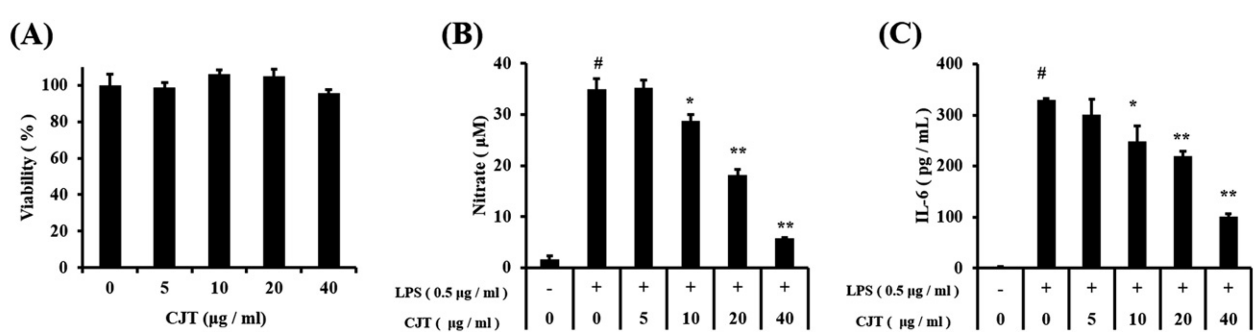


**Fig. 5.** RAW264.7 cells were treated with 5, 10, 20 or 40 μg/ml CJT for 24 h. Survival rates were assessed by MTT assay (A). CJT treatment inhibited NO (B) and IL-6 production (C) in LPS-stimulated RAW264.7 cells. Cells were treated with CJT (5, 10, 20 or 40 μg/ml) for 1 h and incubated with LPS (0.5 μg/ml) for 24 h. The values are expressed as the means ± SD. #Significantly different from the controls, $P < 0.05$; *, **Significantly different from LPS-stimulated cells, $P < 0.05$ and $P < 0.01$.

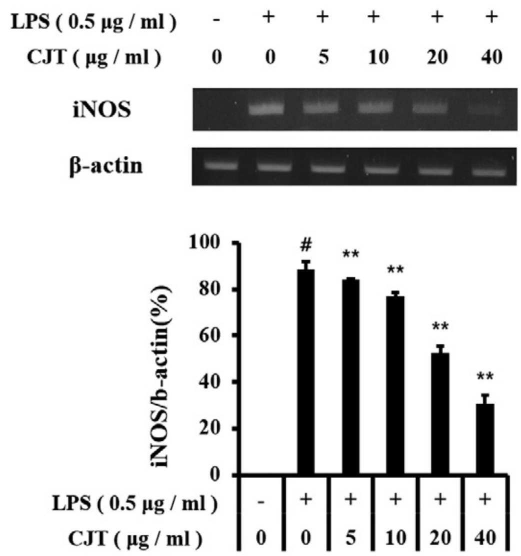


**Fig. 6.** CJT treatment reduced iNOS mRNA levels in LPS-stimulated RAW264.7 cells. β-Actin was used to confirm equal mRNA loading. #Significantly different from the controls, $P < 0.05$; *, **significantly different from LPS-stimulated cells, $P < 0.05$ and $P < 0.01$.

of IL-6, myeloperoxidase and NF-κB. Considering these properties, anti-inflammatory effects of CJT is considered to be closely associated with the effects of actioside and forsythoside B.

In conclusion, the present study demonstrated that pretreatment with CJT significantly attenuated LPS-induced acute lung injury in a mouse model and in LPS-stimulated RAW264.7 cells, which is possibly linked with inhibited production of pro-inflammatory cytokines such as IL-6 and iNOS expression. Histological examination also revealed that CJT treatment reduced anti-inflammatory activity in an LPS-induced ALI mouse model. These results suggest that CJT treatment has an anti-inflammatory effect in acute lung injury.

## Acknowledgments

This work was supported by a grant from the Ministry of Science, ICT and Future Planning (FGC1011433) and the KRIBB Research Initiative Program (KGM1221521) of the Republic of Korea.

## References

[1] Wan LM, Tan L, Wan ZR, Liu SX, Liang SY, Zhong JB, et al. Preventive and therapeutic effects of Danhong injection on lipopolysaccharide induced acute lung injury in mice. J Ethnopharmacol 2013;149:352–9.
[2] Xie X, Sun S, Zhong W, Soromou LW, Wei M, Ren Y, et al. Zingerone attenuates lipopolysaccharide-induced acute lung injury in mice. Int Immunopharmacol 2014;19:103–9.
[3] Tsai CL, Lin YC, Wang HM, Chou TC. Baicalein, an active component of *Scutellaria baicalensis*, protects against lipopolysaccharide-induced acute lung injury in rats. J Ethnopharmacol 2014;153:197–206.
[4] Chen N, Wu Q, Chi G, Soromou LW, Hou J, Deng Y, et al. Prime-O-glucosylcimifugin attenuates lipopolysaccharide-induced acute lung injury in mice. Int Immunopharmacol 2013;16:139–47.
[5] Koarai A, Ichinose M, Sugiura H, Tomaki M, Watanabe M, Yamagata S, et al. iNOS depletion completely diminishes reactive nitrogen-species formation after an allergic response. Eur Respir J 2002;20:609–16.
[6] Kim HG, Yeon SM, Kim KH, Kim H, Park JI, Cha EJ, et al. Estrogenic endocrine-disrupting chemicals modulate the production of inflammatory mediators and cell viability of lipopolysaccharide-stimulated macrophages. Inflammation 2014;1007: 10 [sl0753-014-9966-2].
[7] Kang P, Kim KY, Lee HS, Min SS, Seol GH. Anti-inflammatory effects of anethole in lipopolysaccharide-induced acute lung injury in mice. Life Sci 2013;93:955–61.
[8] Chen J, Wang JB, Yu CH, Chan LQ, Xu P, Yu WY. Total flavonoids of *Mosla scabra* leaves attenuates lipopolysaccharide-induced acute lung injury via down-regulation of inflammatory signaling in mice. J Ethnopharmacol 2013;148:835–41.
[9] Wei D, Huang Z. Anti-inflammatory effects of triptolide in LPS-induced acute lung injury in mice. Inflammation 2014;37:1307–16.
[10] Ono M, Mishima K, Yamasaki T, Masuoka C, Okawa M, Kinjo J, et al. A new lignan glucoside from the stems of *Callicarpa japonica* Thunb. var. *luxurians Rehd*. J Nat Med 2009;63:86–90.
[11] Shin IS, Shin NR, Jeon CM, Hong JM, Kwon OK, Kim JC, et al. Inhibitory effects of Pycnogenol® (French maritime pine bark extract) on airway inflammation in ovalbumin-induced allergic asthma. Food Chem Toxicol 2013;62:681–6.
[12] Yamasaki T, Masuoka C, Nohara T, Ono M. A new phenylethanoid glycoside from the fruits of *Callicarpa japonica* Thunb. var. *luxurians* Rehd. J Nat Med 2007;61: 318–22.
[13] Sung J, Sung M, Kim Y, Ham H, Jeong HS, Lee J. Anti-inflammatory effect of methanol extract from *Erigeron canadensis* L. may be involved with upregulation of heme oxygenase-1 expression and suppression of NFκB and MAPKs activation in macrophages. Nutr Res Pract 2014;8:352–9.
[14] Guan F, Wang H, Shan Y, Wang M, Wang Q, Yin M, et al. Inhibition of COX-2 and PGE2 in LPS-stimulated RAW264.7 cells by lonimacranthoide VI, a chlorogenic acid ester saponin. Biomed Rep 2014;2:760–4.
[15] Koo HJ, Yoon WJ, Sohn EH, Ham YM, Jang SA, Kwon JE, et al. The analgesic and anti-inflammatory effects of *Litsea japonica* fruit are mediated via suppression of NF-κB and JNK/p38 MAPK activation. Int Immunopharmacol 2014;22:84–97.
[16] Fu Y, Zhou E, Wei Z, Wang W, Wang T, Yang Z, et al. Cyanidin-3-O-β-glucoside ameliorates lipopolysaccharide-induced acute lung injury by reducing TLR4 recruitment into lipid rafts. Biochem Pharmacol 2014;90:126–34.
[17] Chen WY, Huang YC, Yang ML, Lee CY, Chen CJ, Yeh CH, et al. Protective effect of rutin on LPS-induced acute lung injury via down-regulation of MIP-2 expression and MMP-9 activation through inhibition of Akt phosphorylation. Int Immuopharmacol 2014;22:409–13.
[18] Zhang Y, Du Z, Zhou Q, Wang Y, Li J. Remifentanil attenuates lipopolysaccharide-induced acute lung injury by downregulating the NF-κB signaling pathway. Inflammation 2014;37:1654–60.
[19] Chen H, Wu S, Lu R, Zhang YG, Zheng Y, Sun J. Pulmonary permeability assessed by fluorescent-labeled dextran instilled intranasally into mice with LPS-induced acute lung injury. PLoS One 2014;9:e101925.
[20] Zhang YM, Zhang SK, Cui NQ. Intravenous infusion of mesenteric lymph from severe intraperitoneal infection rats causes lung injury in healthy rats. World J Gastroenterol 2014;20:4771–7.
[21] Kim S, Lee MS, Lee B, Gwon WG, Joung EJ, Yoon NY, et al. Anti-inflammatory effects of sargachromenol-rich ethanolic extract of *Myagropsis myagroides* on lipopolysaccharide-stimulated BV-2 cells. BMC Complement Altern Med 2014;14:231.
[22] Ayvaz S, Aksu B, Karaca T, Cemek M, Tarladacalisir YT, Ayaz A, et al. Effects of methylene blue in acute lung injury induced by blunt chest trauma. Hippokratia 2014;18: 50–6.
[23] Sun J, Drugan LJ, Zweier JL. Reactive oxygen and nitrogen species regulate inducible nitric oxide synthase function shifting the balance of nitric oxide and superoxide production. Arch Biochem Biophys 2010;494:130–7.
[24] Rentsendorj O, D'Alessio FR, Pearse DB. Phosphodiesterase 2A is a major negative regulator of iNOS expression in lipopolysaccharide-treated mouse alveolar macrophages. J Leukoc Biol 2014;1189:10 [3A0314-152R].
[25] Yingkun N, Zhenyu W, Jing L, Xiuyun L, Huimin Y. Stevioside protects LPS-induced acute lung injury in mice. Inflammation 2013;36:242–50.
[26] Mccluskie K, Birrell MA, Wong S, Belvisi MG. Nitric oxide as a noninvasive biomarker of lipopolysaccharide-induced airway inflammation: possible role in lung neutrophilia. J Phamacol Exp Ther 2004;311:625–33.
[27] Speyer CL, Neff TA, Warner RL, Guo RF, Sarma JV, Riedemann NC, et al. Regulatory effects of iNOS on acute lung inflammatory responses in mice. Am J Pathol 2003; 163:2319–28.
[28] D'Alessio FR, Tsushima K, Aqqarwal NR, Mock JR, Eto Y, Garibaldi BT, et al. Resolution of experimental lung injury by monocyte-derived inducible nitric oxide synthase. J Immunol 2012;189:2234–45.
[29] Wang LF, Mehta S, Weicker S, Scott JA, Joseph M, Razavi HM, et al. Relative contribution of hemopoietic and pulmonary parenchymal cells to lung inducible nitric oxide synthase (iNOS) activity in murine endotoxemia. Biochem Biophys Res Commun 2001;283:694–9.
[30] Mittal M, Siddiqui MR, Tran K, Reddy SP, Malik AB. Reactive oxygen species in inflammation and tissue injury. Antioxid Redox Signal 2014;20:1126–67.
[31] Ravipati AS, Zhang L, Koyyalamudi SR, Jeong SC, Reddy N, Bartlett J, et al. Antioxidant and anti-inflammatory activities of selected Chinese medicinal plants and their relation with antioxidant content. BMC Complement Altern Med 2012;12: 173.
[32] Zhang F, Lu Y, Wu Q, Yan J, Shi J, Liu J. Role of cinnabar and realgar of WSHFD in protecting against LPS-induced neurotoxicity. J Ethnoparmacal 2012;139: 822–8.
[33] Kim JA, Lau EK, Pan L, De Blanco EJ. NF-kappaB inhibitors from *Brucea javanica* exhibiting intracellular effects on reactive oxygen species. Anticancer Res 2010;30: 3295–300.
[34] Saia RS, Mestriner FL, Bertozi G, Cunha FQ, Carnio EC. Cholecystokinin inhibits inducible nitric oxide synthase expression by lipopolysaccharide-stimulated peritoneal macrophages. Mediators Inflamm 2014:896029.
[35] Park JW, Kwon OK, Jang HY, Jeong H, Oh SR, Lee HK, et al. A leaf methanolic extract of *Wercklea insignis* attenuates the lipopolysaccharide-induced inflammatory response by blocking the NF-κB signaling pathway RAW 264.7 macrophages. Inflammation 2012;35:321–31.
[36] Kin SU, Park YH, Min JS, Sun HN, Han YH, Hua JM, et al. Peroxiredoxin I is A ROS/p38 MAPK-dependent inducible antioxidant that regulates NF-κB-mediated iNOS induction and microglial activation. J Neuroimmunol 2013;259:26–36.
[37] Touyz RM, Schiffrin EL. Reactive oxygen species in vascular biology: implication in hypertension. Histochem Cell Biol 2004;122:339–52.

[38] Chen CH, Song TY, Liang YC, Hu ML. Acteoside and 6-O-acetylacteoside downregulate cell adhesion molecules induced by IL-1beta through inhibition of ERK and JNK in human vascular endothelial cells. J Aqric Food Chem 2009;57:8852–9.
[39] Seo ES, Oh BK, Pak JH, Yim SH, Gurunathan S, Kim YP, et al. Acteoside improves survival in cecal ligation and puncture-induced septic mice via blocking of high mobility group box 1 release. Mol Cells 2013;35:348–54.
[40] Jiang WL, Yong-Xu, Zhang SP, Jian-Hou. Forsythoside B protects against experimental sepsis by modulating inflammatory factors. Phytother Res 2012;26:981–7.
[41] Jiang WL, Fu FH, Xu BM, Tian JW, Zhu HB, Jian-Hou. Cardioprotection with forsythoside B in rat myocardial ischemia–reperfusion injury: relation to inflammation response. Phytomedicine 2010;17:635–9.